# Xiao Mao(毛啸)

(Last update: February 17, 2026)

Phone: +1 617 955 7652
Email: matthew99a@gmail.com
xiaomao@stanford.edu
Website: matthew99a.github.io


## Education

- **Stanford University** *2022 to Present*
  Ph.D.
  - Advisor: Aviad Rubinstein
  - Research Area: Theoretical Computer Science
- **Massachusetts Institute of Technology** *2021 to 2022*
  M.Eng.
  - Thesis Supervisor: Virginia Vassilevska Williams
- **Massachusetts Institute of Technology** *2017 to 2021*
  B.S. in Computer Science and Engineering and in Mathematics

## Teaching Experience

- **Design and Analysis of Algorithms** *Fall 2025*
  Course Assistant (Instructor: Mary Wootters)
- **Design and Analysis of Algorithms** *Fall 2024*
  Course Assistant (Instructor: Aviad Rubinstein)
- **Design and Analysis of Algorithms (MIT 6.046J)** *Spring 2022*
  Teaching Assistant
  (Instructors: Srini Devadas, Vinod Vaikuntanathan, Virginia Vassilevska Williams)

## Publications

All authors are listed in lexicographic order.


[1] Xiao Mao and Aviad Rubinstein. Approximation Schemes for Edit Distance and LCS in Quasi-Strongly Subquadratic Time. In *Proceedings of the 58th Annual ACM Symposium on Theory of Computing*, STOC '26, New York, NY, USA, 2026. Association for Computing Machinery

[2] Bartłomiej Dudek, Nick Fischer, Geri Gokaj, Ce Jin, Marvin Künnemann, Xiao Mao, and Mirza Redžić. Classifying Identities: Subcubic Distributivity Checking and Hardness from Arithmetic Progression Detection. In *Proceedings of the 58th Annual ACM Symposium on Theory of Computing*, STOC '26, New York, NY, USA, 2026. Association for Computing Machinery

[3] Ran Duan, Jiayi Mao, Xiao Mao, Xinkai Shu, and Longhui Yin. Breaking the Sorting Barrier for Directed Single-Source Shortest Paths. In *Proceedings of the 57th Annual ACM Symposium on Theory of Computing*, STOC '25, page 36–44, New York, NY, USA, 2025. Association for Computing Machinery. `doi:10.1145/3717823.3718179`
**(STOC 2025 Best Paper) (Invited to the Journal of the ACM)**

[4] Xiao Mao. $(1-\epsilon)$-Approximation of Knapsack in Nearly Quadratic Time. In *Proceedings of the 56th Annual ACM Symposium on Theory of Computing*, STOC 2024, page 295–306, New York, NY, USA, 2024. Association for Computing Machinery. `doi:10.1145/3618260.3649677`

[5] Xiao Mao. Fully Dynamic All-Pairs Shortest Paths: Likely Optimal Worst-Case Update Time. In *Proceedings of the 56th Annual ACM Symposium on Theory of Computing*, STOC 2024, page 1141–1152, New York, NY, USA, 2024. Association for Computing Machinery. `doi:10.1145/3618260.3649695`

[6] Mingyang Deng, Ce Jin, and Xiao Mao. *Approximating Knapsack and Partition via Dense Subset Sums*, pages 2961–2979. URL: `https://epubs.siam.org/doi/abs/10.1137/1.9781611977554.ch113`, `arXiv:https://epubs.siam.org/doi/pdf/10.1137/1.9781611977554.ch113`, `doi:10.1137/1.9781611977554.ch113`

[7] Mingyang Deng, Xiao Mao, and Ziqian Zhong. *On Problems Related to Unbounded SubsetSum: A Unified Combinatorial Approach*, pages 2980–2990. URL: `https://epubs.siam.org/doi/abs/10.1137/1.9781611977554.ch114`, `arXiv:https://epubs.siam.org/doi/pdf/10.1137/1.9781611977554.ch114`, `doi:10.1137/1.9781611977554.ch114`

[8] Xiao Mao. Breaking the Cubic Barrier for (Unweighted) Tree Edit Distance. *SIAM Journal on Computing*, 0(0):FOCS21–195–FOCS21–223, 0. `arXiv:https://doi.org/10.1137/22M1480719`, `doi:10.1137/22M1480719`
**SIAM Journal Version**

[9] Xiao Mao. Breaking the Cubic Barrier for (Unweighted) Tree Edit Distance . In *2021 IEEE 62nd Annual Symposium on Foundations of Computer Science (FOCS)*, pages 792–803, Los Alamitos, CA, USA, February 2022. IEEE Computer Society. URL: `https://doi.ieeecomputersociety.org/10.1109/FOCS52979.2021.00082`, `doi:10.1109/FOCS52979.2021.00082`
**(Machtey Award for Best Student Paper, single author, sole winner) (Published in the SICOMP Special Issue for FOCS 2021)**

## Selected Awards and Scholarships

- **STOC 2025** — *2025*
  Best Paper
- **45th ICPC World Finals** — *November 2022*
  Gold medal, **1st place**
- **FOCS 2021** — *2021*
  Best Student Paper (Machtey Award, single author, sole winner)
- **International Olympiad in Informatics** — *July to August 2017*
  Silver medal, 33th place
- **National Olympiad in Informatics, China** — *July 2016*
  Gold medal, 1st place

## Talks

- **Breaking the sorting barrier for directed single-source shortest paths**
  – Stanford Theory Lunch — *July 2025*
  – Fine-grained Complexity Workshop at EnCORE Institute, UCSD — *May 2025*
  – UCB Theory Lunch — *May 2025*
  – Boston Computation Club — *Jan 2026*

- **Fully Dynamic All-Pairs Shortest Paths: Likely Optimal Worst-Case Update Time**
  – STOC 2024 *June 2024*

- **$(1-\epsilon)$-Approximation of Knapsack in Nearly Quadratic Time**
  – STOC 2024 *June 2024*
  – Stanford Theory Lunch *June 2023*

- **Approximating Knapsack and Partition via Dense Subset Sums**
  – SODA 2023 *Jan 2023*

- **Breaking the Cubic Barrier for (Unweighted) Tree Edit Distance**
  – FOCS 2021 *Feb 2022*
  – Yao Class student seminar *Sep 2021*
  – Theory seminar at the University of Washington *Mar 2022*

## Research and Work Experience

- **Pika (Mellis, Inc.), Palo Alto, CA** *Summer 2025*
  Intern
  - AI research intern.
- **Stanford University** *Sep. 2022 to present*
  Ph.D. currently advised by Prof. Aviad Rubinstein
  - Focus on algorithms and complexity.
- **Massachusetts Institute of Technology** *Sep. 2021 to Sep. 2022*
  M.Eng. with thesis supervised by Prof. Virginia Vassilevska Williams
  - Focus on algorithms and complexity.
- **Massachusetts Institute of Technology** *Feb. 2020 to Dec. 2020*
  UROP advised by Professor Michael Sipser
  - Focus on algorithms and complexity.
- **Microsoft Corporation, Bellevue, WA** *Summer 2019*
  Intern
  - Software engineer intern.
- **Pony.ai, Inc., Fremont, CA** *Summer 2018*
  Intern
  - Software engineer intern.

## Service

- Conference Reviewing: ITCS 2022, SWAT 2022, MFCS 2022, SODA 2024, STOC 2024, FOCS 2024, SODA 2025, SOSA 2025, STOC 2025, SODA 2026, STOC 2026